Circuits RLC / RLC-Schaltungen

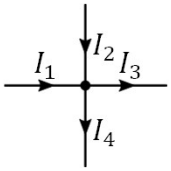
Loi d'Ohm / Ohmsches Gesetz

$$U = I \times R$$

Loi des nœuds / Knotenpunktsatz

$$\sum I = 0$$

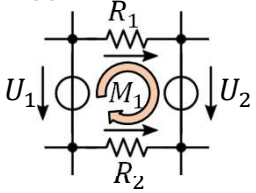
$$I_1 + I_2 - I_3 - I_4 = 0$$



Loi des mailles / Maschensatz

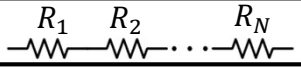
$$\sum U = 0$$

$$U_1 - U_{R1} - U_2 + U_{R2} = 0$$



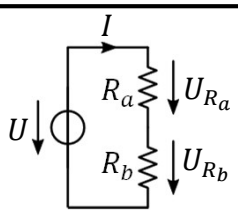
Résistances séries et diviseur de tension / Serienwiderstände und Spannungsteiler

$$R_{EQ} = R_1 + R_2 + \cdots + R_N$$



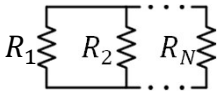
$$U = U_{R_a} + U_{R_b}$$

$$U_{R_a} = U\frac{R_a}{R_a + R_b}, \qquad U_{R_b} = U\frac{R_b}{R_a + R_b}$$

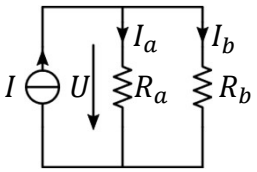


Résistances parallèles et diviseur de courant / Parallelwiderstände und Stromteiler

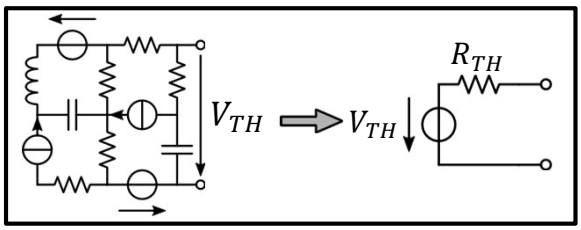
$$R_{EQ} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}}, \qquad R_{EQ} = \frac{R_1 R_2}{R_1 + R_2}$$

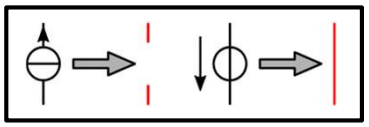


$$I_a = I\frac{R_b}{R_a + R_b}, \qquad I_b = I\frac{R_a}{R_a + R_b}$$



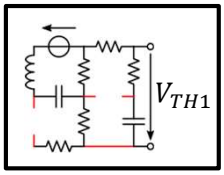
Superposition, circuit équivalent / Superposition, Ersatzschaltbild

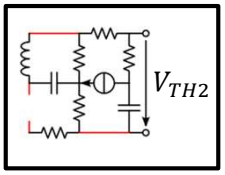


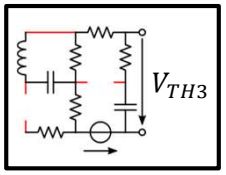


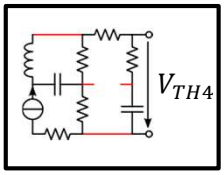
Remplacez chaque source sauf une par un court-circuit (source de tension) ou un circuit ouvert (source de courant)

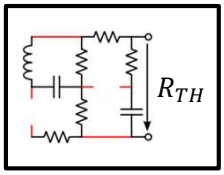
Ersetzen Sie jede Quelle bis auf eine mit einem Kurzschluss (Spannungsquelle) oder einem offenen Stromkreis (Stromquelle).





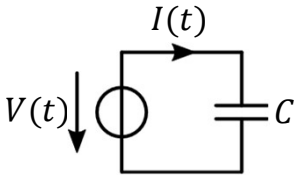




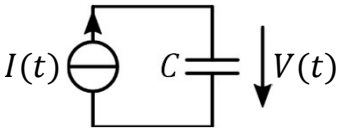


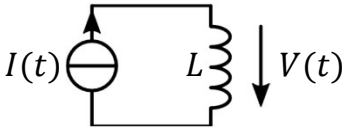
Circuits RLC / RLC-Schaltungen

Condensateurs et Inductances dans le domaine temporel
Kondensatoren und Induktivitäten im Zeitbereich

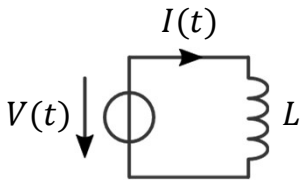


$$I(t) = C\frac{dV(t)}{dt} \qquad V(t) = \frac{1}{C}\int_{t_0}^{t} I(\tau)d\tau + V(t_0)$$





$$V(t) = L\frac{dI(t)}{dt} \qquad I(t) = \frac{1}{L}\int_{t_0}^{t} V(\tau)d\tau + I(t_0)$$

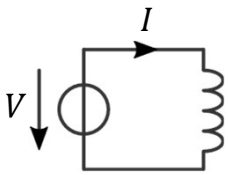


Condensateurs – approximation linéaire
Kondensatoren – lineare Näherung

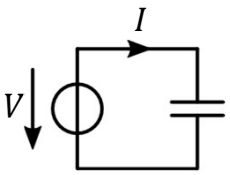
$$\Delta V_C = \frac{I\Delta T}{C} = \frac{\Delta T(V_1 - V_2)}{RC}$$
$$\Delta V_C \ll (V_1 - V_2)$$

$\frac{\Delta V_C}{V_1 - V_2}$ lin	$\frac{\Delta V_C}{V_1 - V_2}$ exp
1%	0.995%
2%	1.98%
5%	4.88%
10%	9.5%
20%	18%
25%	22%

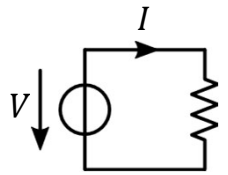
Circuits linéaires complexe – domaine fréquentiel
Komplexe lineare Schaltungen – Frequenzbereich



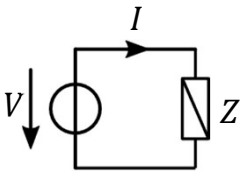
$$Z_L = j\omega L = \omega L\, e^{j\frac{\pi}{2}}$$



$$Z_C = \frac{1}{j\omega C} = \frac{1}{\omega C}\, e^{-j\frac{\pi}{2}}$$



$$Z_R = R$$



Général // Allgemein